

Customer Churn

Exploratory Data Analysis Report

Telco Customer Churn Dataset | Source: Kaggle.com

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Tools: Python · Pandas · NumPy · Seaborn · Matplotlib · VS Code

7,043

Total Records

21

Features

26.5%

Churn Rate

3

Top Factors

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SECTION 1

Introduction & Objectives

Customer churn refers to when customers stop using a company's products or services. For telecom companies, churn is a critical metric — acquiring a new customer costs 5–7× more than retaining an existing one. Even a small improvement in retention directly boosts revenue and customer lifetime value.

This report analyses the Telco Customer Churn dataset (Kaggle) using Python-based EDA techniques to understand churn patterns, identify key drivers, and suggest actionable business strategies.

Objectives

- Understand the structure, distribution, and quality of the dataset.
- Identify features most strongly correlated with customer churn.
- Visualise patterns using count plots, pie charts, histograms, boxplots, heatmaps and scatter plots.
- Derive key findings and translate them into business recommendations.

SECTION 2

Dataset Details

Source: **Kaggle.com — Telco Customer Churn Dataset**. The dataset represents a US telecom company's customer base with demographics, services, account information, and churn status.

Attribute		Detail
Total Records		7,043 customers
Total Features		21 columns (20 inputs + 1 target)
Target Variable		Churn — Yes (churned) / No (retained)
Churn Rate		~26.5% of customers churned
ID Column		customerID — dropped (not useful for analysis)

Feature	Type	Description
tenure	Numerical	Months the customer has been with the company
MonthlyCharges	Numerical	Monthly fee billed to the customer
TotalCharges	Numerical	Total amount billed (had data type issue)
Contract	Categorical	Month-to-month / One year / Two year
PaymentMethod	Categorical	Electronic check, Bank transfer, Credit card, etc.
InternetService	Categorical	DSL / Fiber optic / No
SeniorCitizen	Categorical	1 = Senior citizen, 0 = Non-senior
Churn (Target)	Categorical	Yes = churned, No = retained

SECTION 3

Data Cleaning Summary

The dataset was cleaned before analysis to ensure accuracy and consistency. The following steps were performed:

Cleaning Step	Action Taken
Missing Values	isnull().sum() check. TotalCharges had blank strings → converted using pd.to_numeric(errors='coerce')
Duplicates	duplicated().sum() — 0 duplicate rows found.
Data Type Fix	TotalCharges was object (string) → corrected to float64.
Column Removal	customerID dropped — serves only as a unique identifier.
Whitespace	All string columns stripped with lambda x: x.str.strip().
Unique Value Check	Contract column verified with unique() to detect inconsistencies.
Outlier Detection	IQR method on MonthlyCharges: lower = $Q1 - 1.5 \times IQR$, upper = $Q3 + 1.5 \times IQR$ → 0 outliers found.

Data Quality Summary After Cleaning

Issue	Column(s)	Resolution
Missing Values	TotalCharges	11 blank strings coerced to NaN, then handled
Duplicates	All columns	None found
Wrong Data Type	TotalCharges	object → float64 corrected
Outliers	MonthlyCharges	0 outliers — IQR method confirmed clean data
Whitespace	All object cols	Stripped using str.strip()

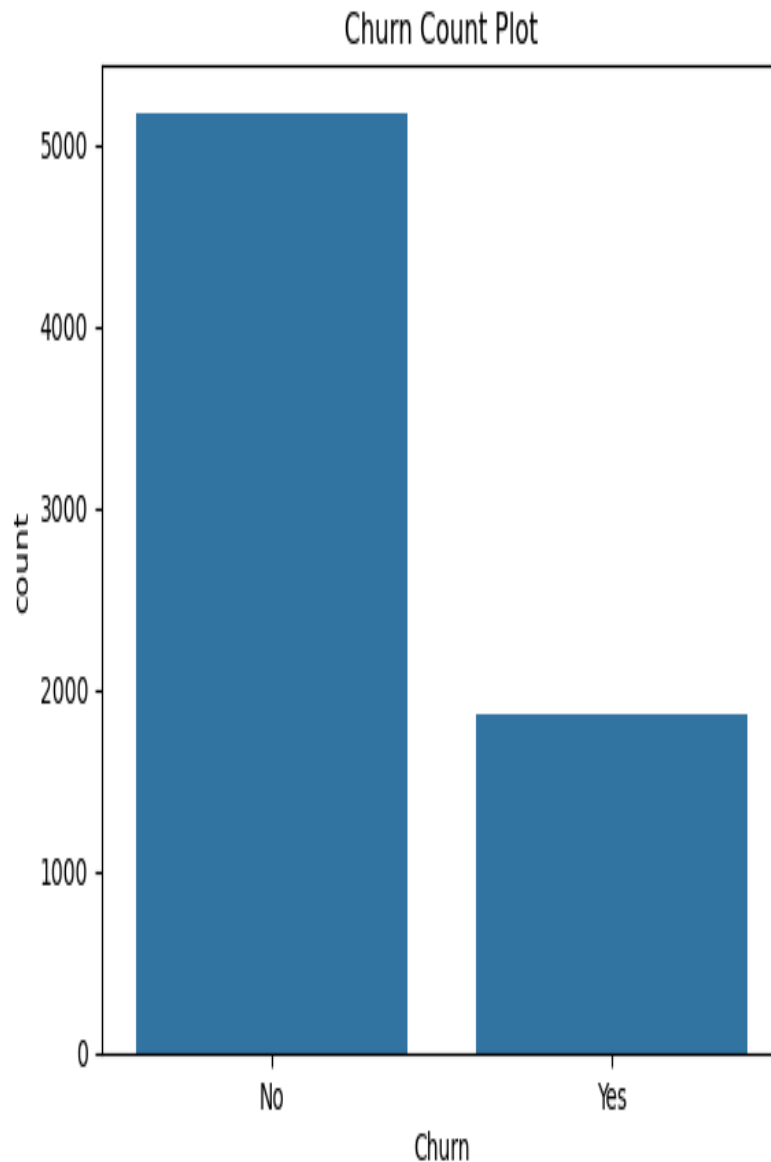
SECTION 4

Exploratory Data Analysis

4.1 Univariate Analysis

Univariate analysis studies one variable at a time to understand its distribution.

Chart 1 — Churn Count Plot

**Observation:**

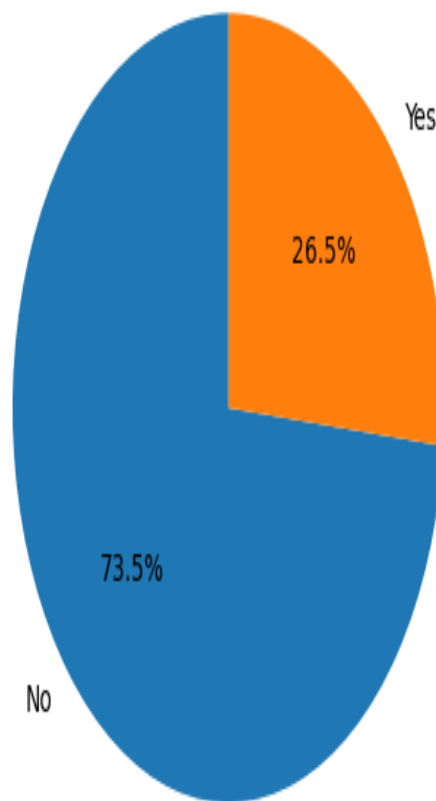
5,174 customers did NOT churn (~73.5%) and 1,869 DID churn (~26.5%). Clear class imbalance — retained customers outnumber churned ones nearly 3:1.

Business Meaning:

A 26.5% churn rate is above the telecom industry average. Even retaining an extra 5% of churners would recover significant monthly recurring revenue.

Chart 2 — Churn Percentage (Pie Chart)

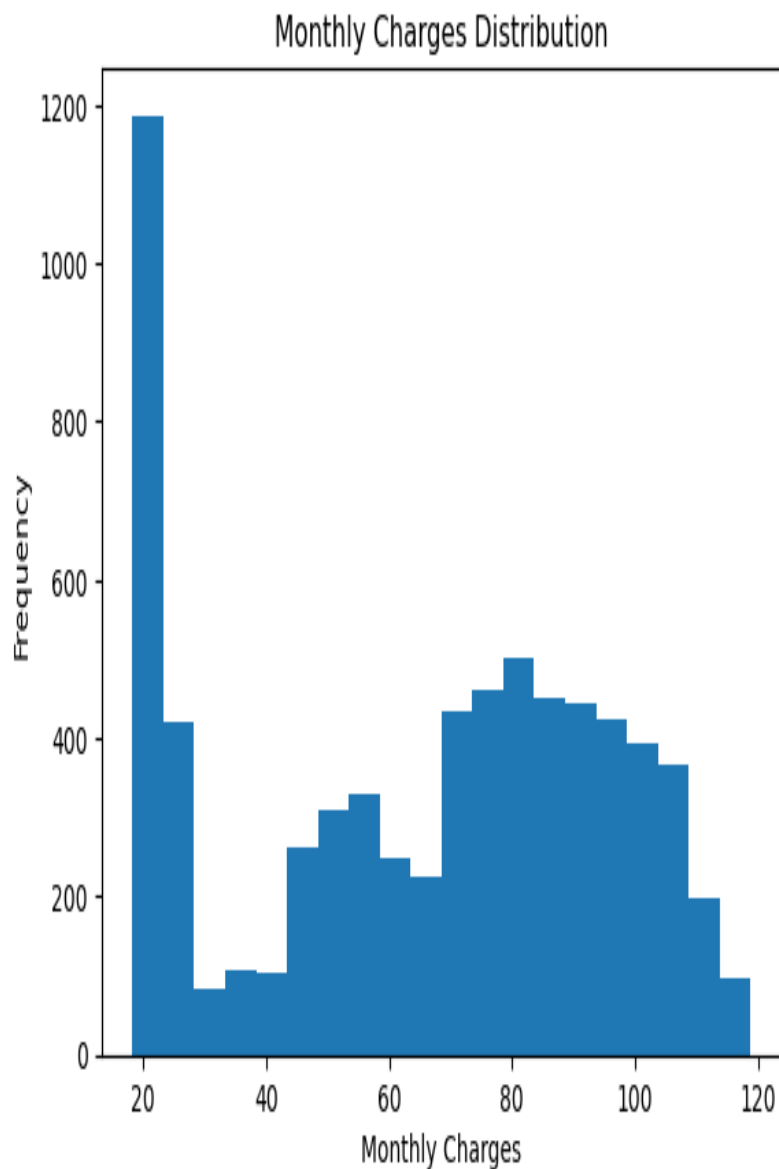
Churn Percentage

**Observation:**

73.5% retained (No) vs 26.5% churned (Yes). Roughly 1 in 4 customers leaves.

Business Meaning:

This proportion signals an urgent need for targeted retention programmes, particularly for high-risk segments identified in the bivariate analysis.

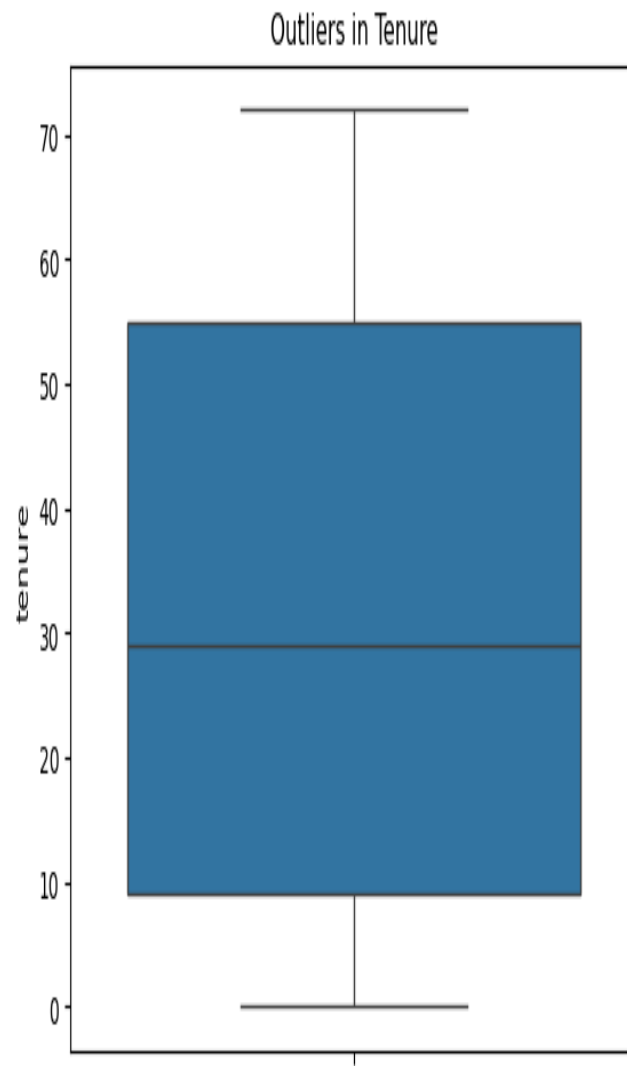
Chart 3 — Monthly Charges Distribution**Observation:**

Bimodal distribution: large cluster around \$20/month (basic plans) and another around \$70–\$90/month (premium plans). Fewer customers sit in the middle range.

Business Meaning:

Customers on expensive premium plans may feel the cost is not justified — especially new customers. The company should communicate value clearly and offer flexible mid-range options to reduce price-driven churn.

Chart 4 — Outliers in Tenure Distribution (Boxplot)

**Observation:**

Tenure ranges from 0 to 72 months, median ~29 months. No extreme outliers detected. Distribution covers all lifecycle stages — new joiners to long-term loyal customers.

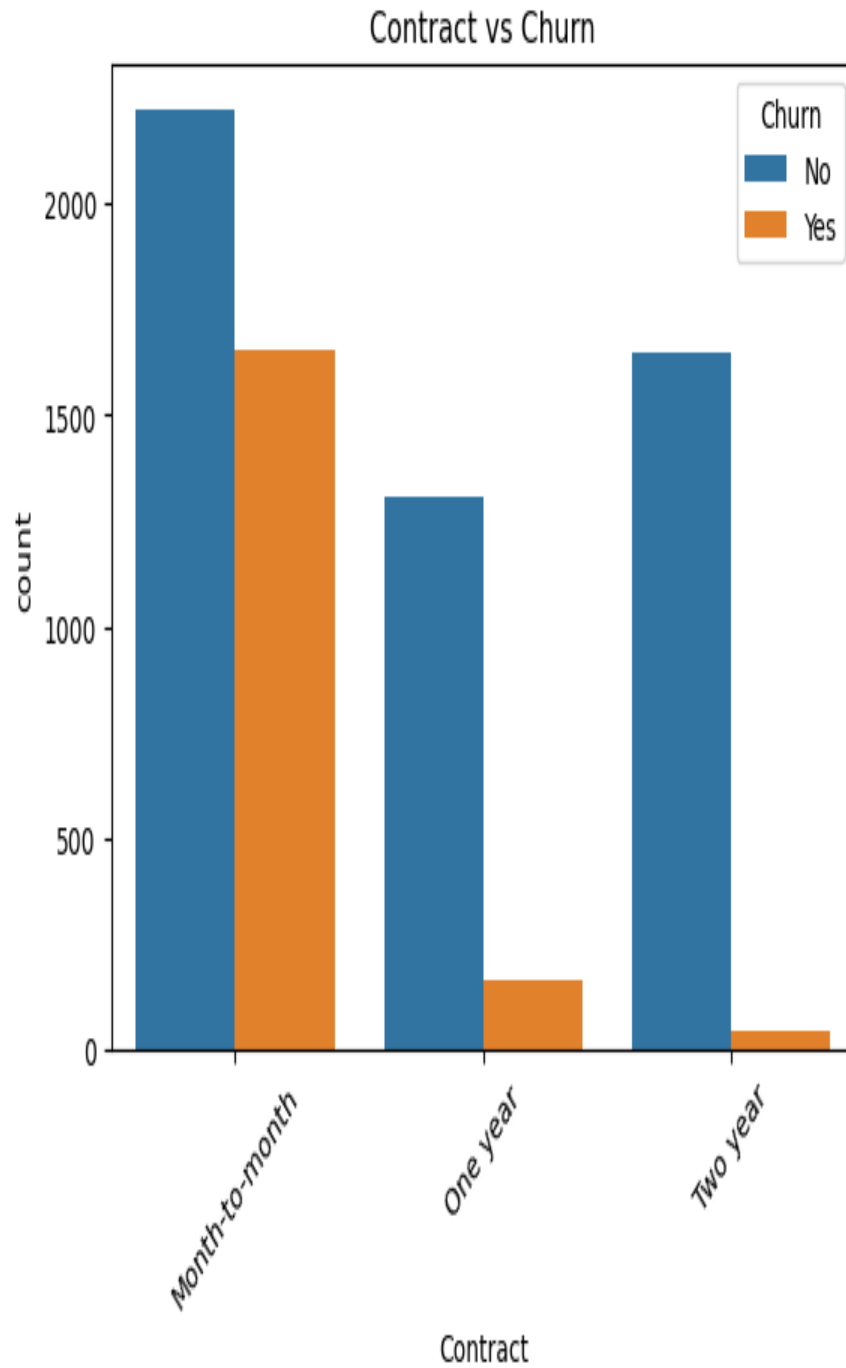
Business Meaning:

Low-tenure customers (0–12 months) are the highest churn risk group. First-year engagement and onboarding programmes are critical to build loyalty.

4.2 Bivariate Analysis

Bivariate analysis examines the relationship between two variables — specifically how each feature relates to the Churn outcome.

Chart 5 — Contract Type vs Churn



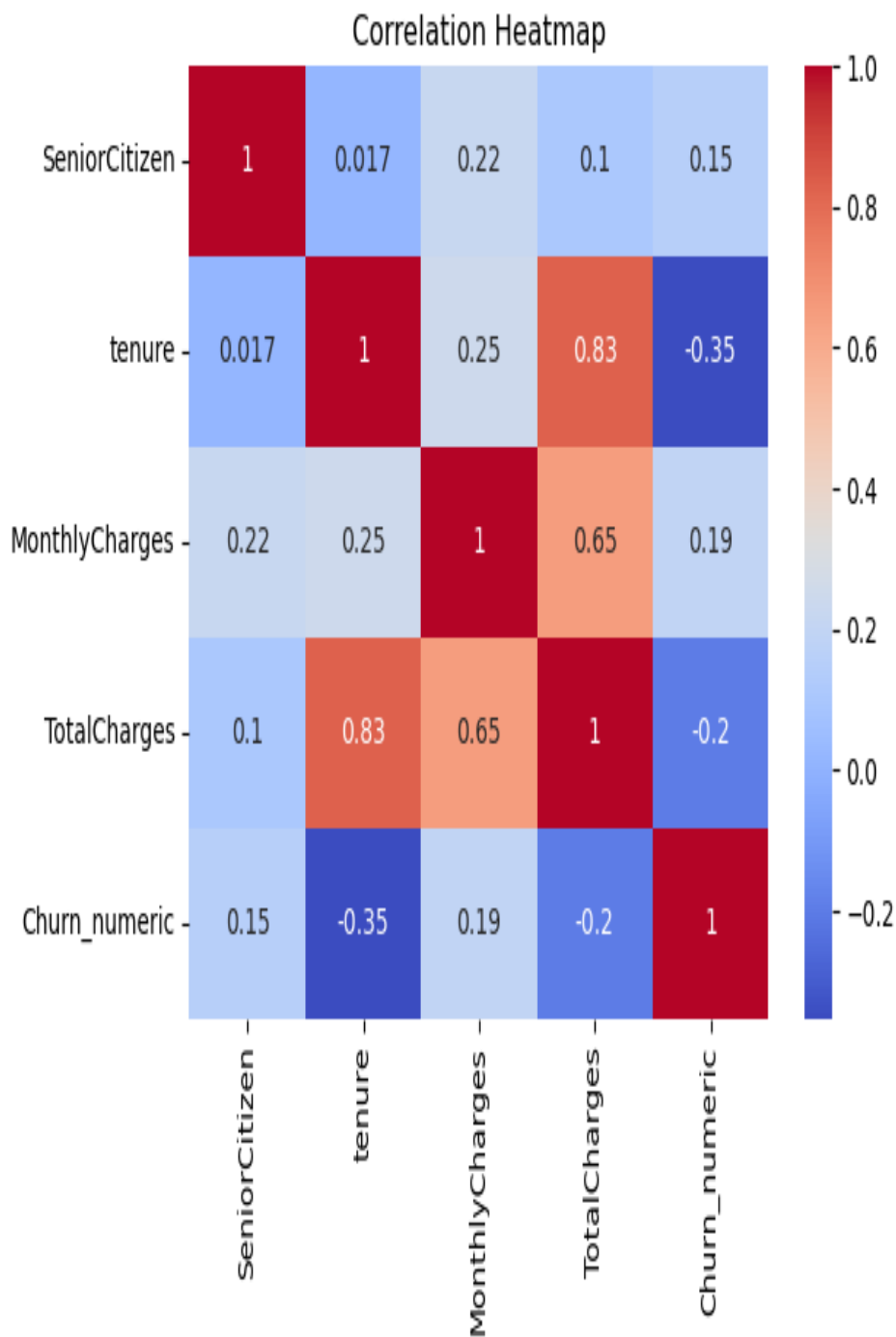
Observation:

Month-to-month contract customers have by far the highest churn. One-year contract customers churn moderately. Two-year contract holders show almost zero churn — they are the most loyal segment.

Business Meaning:

Contract type is the single strongest churn predictor in this dataset. Customers on flexible contracts have no switching cost. Offering meaningful incentives (10–20% discounts, free service months) to commit to 1–2 year plans can dramatically reduce churn.

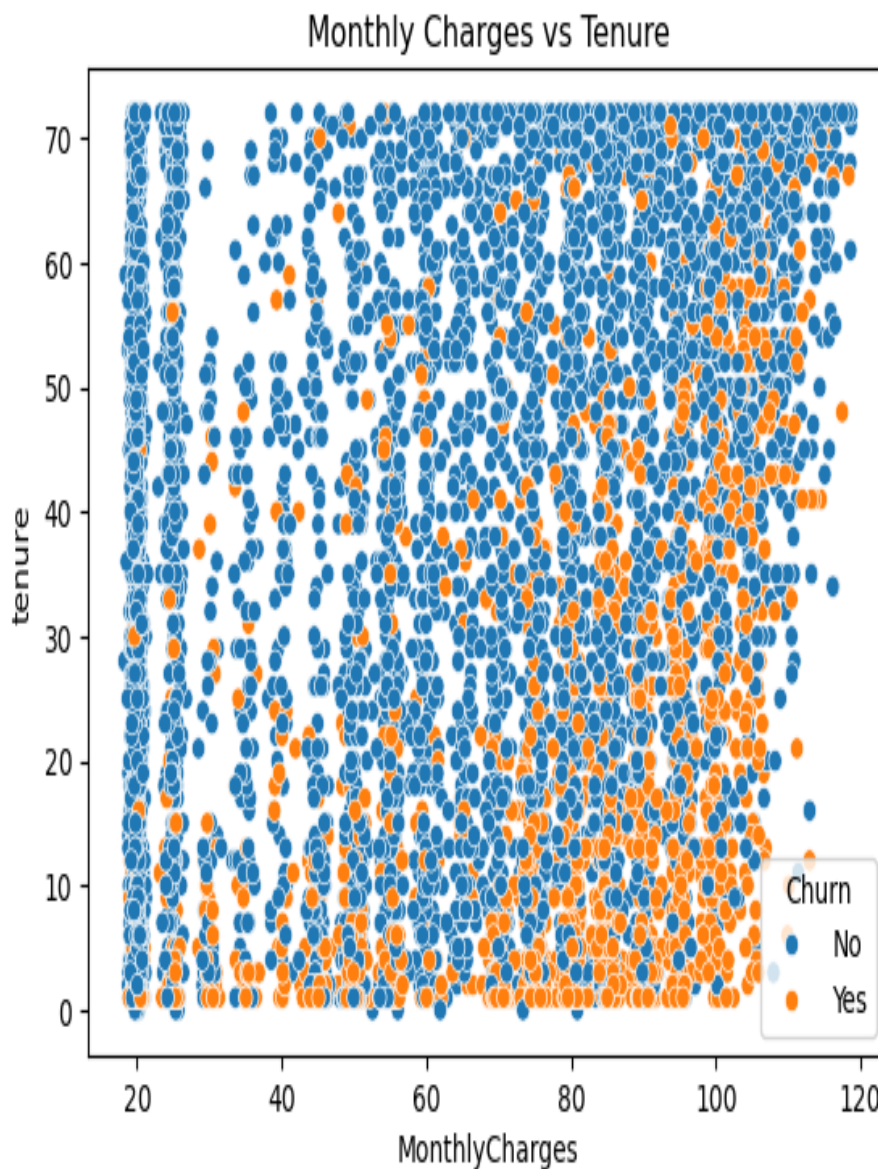
Chart 6 — Correlation Heatmap

**Observation:**

Tenure has a notable negative correlation with churn — longer-tenure customers rarely leave. MonthlyCharges shows a positive correlation with churn — higher fees link to more churn. TotalCharges correlates strongly with tenure (it accumulates over time).

Business Meaning:

Tenure and MonthlyCharges are the two key numerical churn drivers. Any future machine learning model should prioritise these features. Marketing should segment customers with high charges and low tenure as the primary at-risk group and target them with personalised retention offers.

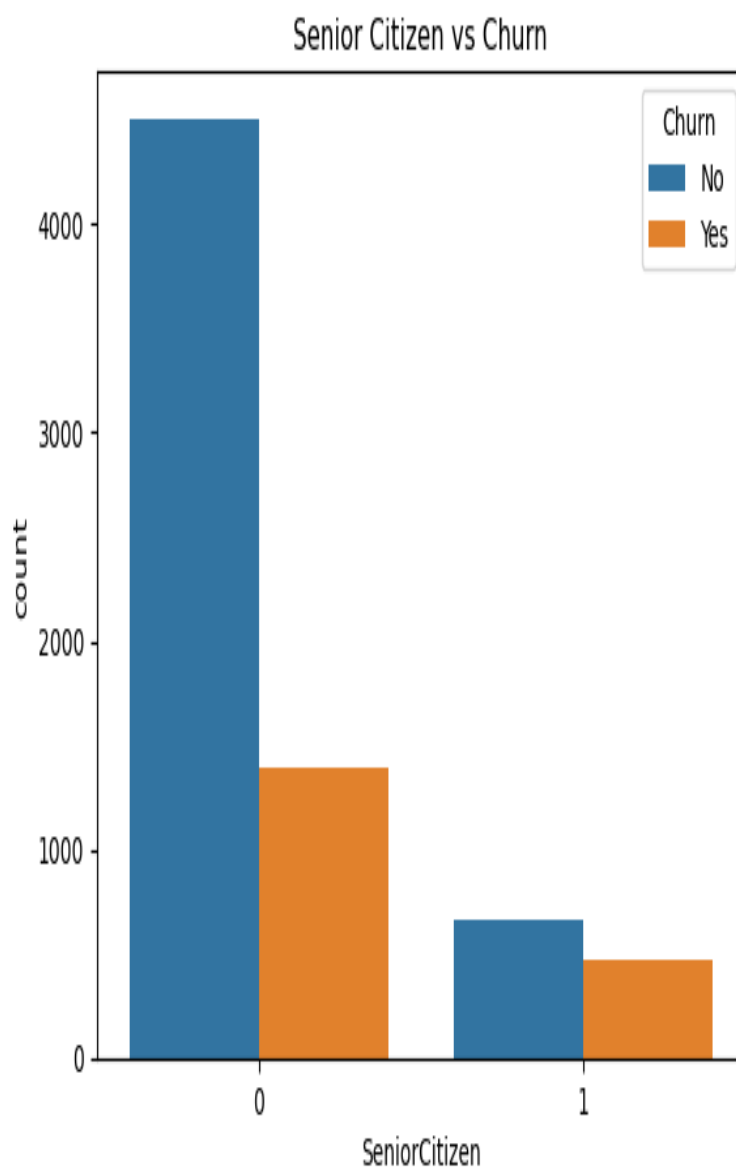
Chart 7 — Monthly Charges vs Tenure (Scatter Plot)**Observation:**

Churned customers cluster heavily in the bottom-left — low tenure AND high monthly charges. Retained customers dominate the right side (high tenure). This is the clearest visual separation between churned and retained customers.

Business Meaning:

New customers paying premium prices are the most vulnerable. This points to a value perception gap in the early customer lifecycle. Better onboarding, transparent billing, and early-stage discounts can reduce this.

Chart 8 — Senior Citizen vs Churn

**Observation:**

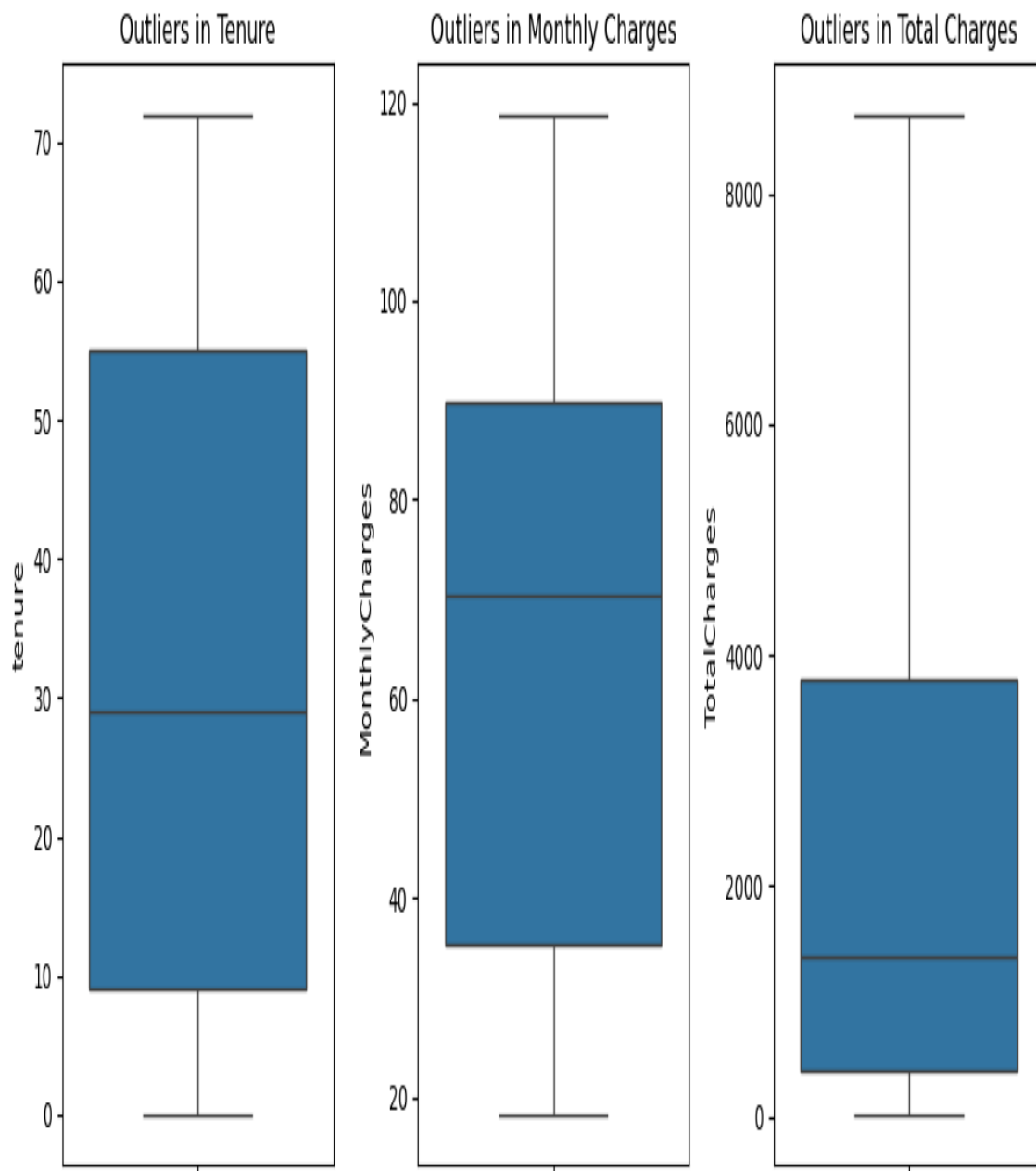
Senior citizens (SeniorCitizen = 1) are a small share of customers but show a proportionally higher churn rate than non-seniors. Among seniors, churn is visibly more frequent relative to their total population.

Business Meaning:

Senior customers may struggle with complex offerings, digital billing, or service changes. Dedicated support lines, simplified plan options, and personalised outreach can meaningfully reduce churn in this demographic.

4.3 Outlier Analysis

Chart 9 — Outlier Analysis: Tenure · Monthly Charges · Total Charges



Observation:

Three-panel boxplot: (1) Tenure — 0–72 months, median ~29, no outliers. (2) MonthlyCharges — \$18–\$118, uniform spread, no extreme outliers. (3) TotalCharges — wider spread due to tenure × charge compounding, also clean.

Business Meaning:

Zero outliers across all three key numerical features confirms strong data quality. The cleaned dataset is reliable for statistical analysis and machine learning model development — no outlier removal is needed.

EDA Questions & Answers Summary

Question	Answer
What % of customers churned?	~26.5% churned; ~73.5% retained
Highest churn group?	Month-to-month + high charges + low tenure
Top numerical churn features?	Tenure (negative) and MonthlyCharges (positive)
Top categorical churn features?	Contract type, PaymentMethod, InternetService, SeniorCitizen
Missing values / data quality issues?	TotalCharges had type error — fixed; no major issues
Outliers in dataset?	0 outliers — IQR method confirmed clean data
Top 3 churn factors?	Contract type, Tenure, Monthly Charges

SECTION 5

Key Findings

The following are the most important churn patterns discovered through this analysis:

F1 Month-to-Month Contracts Drive the Highest Churn

Customers on monthly contracts churn at the highest rate — they face no switching cost and can leave at any time. This is the single most powerful churn predictor in the dataset.

F2 Low Tenure = High Churn Risk

Customers in their first 0–12 months are the most vulnerable. If early-stage loyalty is not built, long-term revenue suffers. The first year is the most critical retention window.

F3 High Monthly Charges Correlate with Churn

Customers paying \$70+ per month churn more often. Price sensitivity is a major driver, especially among newer customers still evaluating whether the service is worth the cost.

F4 Payment Method Influences Churn

Electronic check users show disproportionately higher churn. Payment friction or lack of trust in the payment process may be contributing to this behaviour.

F5 Senior Citizens Are a Higher-Risk Demographic

Though a minority, senior citizens churn proportionally more — likely due to difficulty navigating complex digital services and plan options.

F6 Dataset Is Clean and Ready for Machine Learning

Zero outliers, zero duplicates, and only minor type corrections needed. This dataset is well-suited for building a churn prediction model as the next analytical step.

SECTION 6

Business Recommendations

Based on the findings above, the following evidence-based recommendations are proposed specifically for the Telco company to reduce churn:

R1 Improve New Customer Onboarding (Months 0–12)

Customers with low tenure churn most. Design a structured first-year journey: welcome call on Day 1, setup guide by Day 7, check-in offer at Month 3, and a loyalty reward at Month 6. Making customers feel valued early reduces early churn.

R2 Incentivise Long-Term Contract Commitments

Month-to-month customers are the highest churn risk. Offer 10–20% discounts, free service months, or data/speed upgrades for customers who commit to 1-year or 2-year plans. This creates switching cost and massively improves retention.

R3 Introduce Value-Based & Flexible Pricing Plans

High monthly charges drive churn among price-sensitive customers. Introduce tiered pricing with a strong mid-range option, bundled deals that clearly communicate value, and loyalty pricing for customers who stay beyond 12 months.

R4 Deploy Proactive Retention for At-Risk Customers

Build a customer health score using tenure, charges, and contract type. Automatically trigger retention calls or discount offers when a customer scores high-risk — before they make the decision to leave.

R5 Simplify the Payment Experience

Reduce electronic check churn by promoting auto-pay (bank transfer / credit card) with a small monthly discount (e.g. \$2–\$5 off). Clear, itemised billing statements will build trust and reduce payment-related frustrations.

R6 Create Senior-Friendly Plans & Dedicated Support

Design simplified plan options for senior citizens — fewer choices, larger print billing, and a dedicated support line. A senior loyalty scheme with assistive features (e.g. tech support, home visits) can significantly improve retention.

R7 Build a Churn Prediction Model as the Next Step

The five features identified — contract type, tenure, MonthlyCharges, PaymentMethod, SeniorCitizen — are strong ML inputs. A logistic regression or gradient boosting classifier on this cleaned dataset will allow the company to score every customer's churn probability in real time and proactively target retention efforts.

SECTION 7

Conclusion

This Exploratory Data Analysis of the Telco Customer Churn dataset provided a thorough understanding of who churns, why they churn, and what can be done about it. Through systematic data cleaning, univariate analysis, bivariate analysis, and correlation study, the following core insights were established:

Factor	Key Finding	Recommended Action
Contract Type	Month-to-month customers churn at highest rate	Incentivise long-term plan migration
Tenure	New customers (0–12 months) are most at risk	Structured onboarding & first-year loyalty
Monthly Charges	Higher charges strongly correlate with churn	Flexible, value-based pricing tiers
Payment Method	Electronic check users show elevated churn	Promote auto-pay with discount
Senior Citizens	Proportionally higher churn in senior segment	Simplified plans + dedicated support
Data Quality	Zero outliers, no duplicates after cleaning	Dataset ready for ML model development

The analysis confirms that churn follows clear, predictable patterns tied to contract flexibility, early-stage customer experience, and pricing. These are all areas the business can directly control and improve. Churn is not inevitable — it is manageable with the right data-driven strategy.

This EDA serves as both a standalone insight report and the analytical foundation for the next logical step: building a supervised machine learning churn prediction model. The features identified here — contract type, tenure, MonthlyCharges, PaymentMethod, and SeniorCitizen status — are strong and well-validated model inputs.

By acting on the seven business recommendations in this report — especially improving onboarding, promoting long-term contracts, and addressing pricing pressure — the company can realistically aim to reduce its 26.5% churn rate by 5–8 percentage points. That reduction alone, on a 7,043-customer base, translates to retaining 350–560 additional customers per cycle, each contributing monthly recurring revenue.

End of Report · Customer Churn EDA · Prepared by Nandini · Telco Dataset — Kaggle